Finding Vertical and Horizontal Asymptotes of Rational Functions

Algebra II Worksheet · Grade 10-12

Name: ______________________ Date: ______________ Score: / 10

Learning Objectives

- Identify and find vertical asymptotes by setting the denominator equal to zero
- Determine horizontal asymptotes by comparing degrees of numerator and denominator
- Graph rational functions using their asymptotes as guides

Find all vertical and horizontal asymptotes of each rational function below, showing your work.

1. Find the vertical asymptote(s) of the rational function.

$$f(x) = \frac{2x^2}{x^2 - 1}$$

Answer: ______________________

2. Find the vertical asymptote(s) of the rational function.

$$f(x) = \frac{x^2 + x - 2}{x^2 - x - 6}$$

Answer: ______________________

3. Find the vertical asymptote(s) of the rational function.

$$f(x) = \frac{x + 5}{x^2 - 9}$$

Answer: ______________________

4. Find the vertical asymptote(s) of the rational function.

$$f(x) = \frac{3x}{x^2 - 4x + 4}$$

Answer: ______________________

5. Find the horizontal asymptote of the rational function.

$$f(x) = \frac{2x^2}{x^2 - 1}$$

Answer: ______________________

6. Find the horizontal asymptote of the rational function.

$$f(x) = \frac{4x + 1}{2x^2 + 3}$$

Answer: ______________________

7. Find the horizontal asymptote of the rational function.

$$f(x) = \frac{5x^3 + 2x}{2x^3 - x + 7}$$

Answer: ______________________________________

8. Find all vertical and horizontal asymptotes of the rational function.

$$f(x) = \frac{x^2 - 4}{x^2 - 5x + 6}$$

Answer: ______________________________________

9. Find all vertical and horizontal asymptotes of the rational function.

$$f(x) = \frac{3x^2 + 1}{x^2 - 16}$$

Answer: ______________________________________

10. Find all vertical and horizontal asymptotes of the rational function.

$$f(x) = \frac{x + 2}{x^2 + x - 12}$$

Answer: ______________________________________

Remind students to factor denominators completely and to check for common factors that would create holes rather than asymptotes.

Solutions

1. Find the vertical asymptote(s) of the rational function.

$$f(x) = \frac{2x^2}{x^2 - 1}$$

→ *Set the denominator equal to zero*
→ *Factor the denominator as a difference of two squares*
→ *Apply the zero product property to each factor*
→ *Solve each linear equation for x*

Answer: $x = -1,\ x = 1$

2. Find the vertical asymptote(s) of the rational function.

$$f(x) = \frac{x^2 + x - 2}{x^2 - x - 6}$$

→ *Set the denominator equal to zero*
→ *Factor the quadratic in the denominator*
→ *Apply the zero product property*
→ *Solve each factor for x*

Answer: $x = -2,\ x = 3$

3. Find the vertical asymptote(s) of the rational function.

$$f(x) = \frac{x + 5}{x^2 - 9}$$

→ *Set the denominator equal to zero*
→ *Factor as a difference of two squares*
→ *Use the zero product property*
→ *Solve for both values of x*

Answer: $x = -3,\ x = 3$

4. Find the vertical asymptote(s) of the rational function.

$$f(x) = \frac{3x}{x^2 - 4x + 4}$$

→ *Set the denominator equal to zero*
→ *Factor the perfect square trinomial*
→ *Apply the zero product property*
→ *Solve for x to find one repeated vertical asymptote*

Answer: $x = 2$

5. Find the horizontal asymptote of the rational function.

$$f(x) = \frac{2x^2}{x^2 - 1}$$

→ *Compare the degrees of the numerator and denominator*
→ *Both degrees are equal, so divide leading coefficients*
→ *The horizontal asymptote is the ratio of the leading coefficients*

Answer: $y = 2$

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6. Find the horizontal asymptote of the rational function.

$$f(x) = \frac{4x + 1}{2x^2 + 3}$$

→ *Compare the degree of the numerator and denominator*
→ *The numerator degree is less than the denominator degree*
→ *When the bottom degree is larger, the horizontal asymptote is y equals zero*

Answer: $y = 0$

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7. Find the horizontal asymptote of the rational function.

$$f(x) = \frac{5x^3 + 2x}{2x^3 - x + 7}$$

→ *Compare the degrees of numerator and denominator*
→ *Both have degree three so they are equal*
→ *Divide the leading coefficient of the numerator by that of the denominator*

Answer: $y = \frac{5}{2}$

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8. Find all vertical and horizontal asymptotes of the rational function.

$$f(x) = \frac{x^2 - 4}{x^2 - 5x + 6}$$

→ *Set the denominator equal to zero and factor*
→ *Solve to find vertical asymptotes at x equals 2 and x equals 3*
→ *Compare degrees of numerator and denominator*
→ *Since degrees are equal divide leading coefficients to find the horizontal asymptote*

Answer: $\text{Vertical: } x = 2,\ x = 3;\ \text{Horizontal: } y = 1$

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9. Find all vertical and horizontal asymptotes of the rational function.

$$f(x) = \frac{3x^2 + 1}{x^2 - 16}$$

→ *Factor the denominator as a difference of two squares*
→ *Set each factor equal to zero and solve*
→ *Compare degrees of numerator and denominator*
→ *Both degrees are two so divide leading coefficients for the horizontal asymptote*

Answer: $\text{Vertical: } x = -4,\ x = 4;\ \text{Horizontal: } y = 3$

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10. Find all vertical and horizontal asymptotes of the rational function.

$$f(x) = \frac{x + 2}{x^2 + x - 12}$$

→ *Factor the denominator*
→ *Set each factor equal to zero and solve for x*
→ *Compare numerator and denominator degrees*
→ *Numerator degree is smaller so the horizontal asymptote is y equals zero*

Answer: Vertical: $x = -4,\ x = 3;$ Horizontal: $y = 0$